Heat and Work
26 Questions Explained

热与功：26 道选择题
中英文对照讲解

Original English questions and all four choices, with complete Chinese translations, worked mathematics, and explanations of the incorrect alternatives.

保留英文原题与全部四个选项，配完整中文翻译、公式推导及错误选项辨析。

Source quiz: *2 Quiz heat-work multiple-choice*, in *Thermodynamics, Ch. 1,2,3*. All 26 questions were read from the currently displayed quiz and its revealed feedback.

来源为当前页面的“2 Quiz heat-work multiple-choice”。已读取全部 26 道题及展开后的答案反馈，并按原顺序整理。

Editorial checks distinguish the quiz's intended answer from mathematical correctness. In particular, Questions 2, 3, and 8 have overlapping or equivalent choices; Questions 19 and 20 need an explicit work-sign convention.

本讲义区分“原题预期答案”与“严格数学结论”。第 2、3、8 题的选项存在重叠或等价问题；第 19、20 题必须明确输出功与对系统做功的符号。

Prepared for Jiasheng Fu 29 October 2026

题干、选项、讲解和易错点均提供中英对照；公式采用分式、积分和分行推导。

Answer key / 答案总表

Q	Key	Q	Key	Q	Key	Q	Key
1	C	8	A*	15	C	22	B
2	D*	9	C	16	D*	23	A
3	B*	10	B	17	C	24	A
4	A	11	C	18	B	25	B
5	B	12	D	19	C*	26	B
6	B	13	A	20	D*		
7	B	14	A	21	C*		

The table gives the quiz's marked answers. An asterisk identifies an important qualification or correction in the explanation. For Q3, A and B are mathematically identical; for Q8, C is also a true identity.

表中列出原题标记的答案。星号表示讲解中有重要条件或纠正。第 3 题 A、B 数学上完全相同；第 8 题 C 也是正确恒等式。

Sign convention / 符号约定

Unless stated otherwise, $q > 0$ means heat enters the system, and $w > 0$ means work is done on it. Positive work output is denoted $W_{\text{out}} = -w$.

除非另有说明，$q > 0$ 表示系统吸热，$w > 0$ 表示外界对系统做功。若把对外输出功记为正值，则 $W_{\text{out}} = -w$。

$$\boxed{\Delta U = q + w}, \qquad \delta w_{pV} = -p_{\text{ex}}\,\mathrm{d}V.$$

Scope and notation / 适用范围与记号

The basic energy balances concern closed systems with negligible changes in bulk kinetic and potential energies. Formula-specific assumptions are stated beside each derivation. C is total heat capacity; C_{m} is molar heat capacity.

基本能量平衡针对封闭系统，并忽略整体动能、势能变化。每个公式的额外适用条件在推导旁说明。C 表示总热容，C_{m} 表示摩尔热容。

Standard-state $\circ$ and the quiz's $\ominus$ notation are used equivalently. ln is the natural logarithm. Temperatures in gas laws and Boltzmann factors are kelvin.

标准态符号 $\circ$ 与原题使用的 $\ominus$ 在此等价；ln 为自然对数。气体方程、玻尔兹曼因子中的温度均用开尔文。

Reading and navigation / 阅读与导航

Each question starts on a new page. PDF bookmarks jump directly to questions. The English source wording is preserved, including flawed claims in options; corrections appear in the explanation.

每题另起一页，可通过 PDF 书签跳转。英文题干和选项保留原文，包括其中不严谨的陈述；相应纠正在讲解中明确指出。

QUESTION 1 / 26

Maximum work in isothermal expansion

等温膨胀的最大功

Which statement correctly describes the work done when a perfect gas expands isothermally from an initial volume V_i to a final volume V_f?

理想气体从初始体积 V_i 等温膨胀到最终体积 V_f 时，下列哪项正确描述了所做的功？

A. Reversible isothermal expansion results in no net transfer of heat into the system.

可逆等温膨胀过程中，没有净热量传入系统。

B. The reversible work done is given by $w = -p_{\text{ex}}\Delta V$ where p_{ex} remains constant.

可逆过程的功为 $w = -p_{\text{ex}}\Delta V$，且外压 p_{ex} 保持不变。

C. The work done by the gas is maximum when the expansion is carried out reversibly because p_{ex} is matched to p at every stage.

可逆膨胀时，外压 p_{ex} 在每一步都与气体压强 p 相匹配，因此气体对外做功最大。

D. The work done during irreversible expansion against a constant external pressure is always greater than during reversible expansion.

对抗恒定外压的不可逆膨胀，其对外做功总比可逆膨胀大。

Answer and reasoning / 答案与推导

C

For fixed equilibrium endpoints and a common isothermal reservoir, a reversible expansion delivers the maximum work output. In the chemistry convention, expansion work on the gas is negative. Since an ideal gas has $\Delta U = 0$ at fixed T, absorbed heat exactly compensates the work output.

在相同平衡始末态和同一恒温热库下，可逆膨胀的对外输出功最大。按化学符号约定，膨胀时外界对气体做功为负。理想气体等温时 $\Delta U = 0$，所以吸热量恰好补偿对外做功。

$$w_{\text{rev}} = -\int_{V_i}^{V_f} p\,\mathrm{d}V = -nRT\ln\left(\frac{V_f}{V_i}\right),$$
$$q_{\text{rev}} = -w_{\text{rev}} > 0, \qquad W_{\text{out,max}} = -w_{\text{rev}}.$$

Alternatives and conditions / 选项辨析与适用条件

A omits the required heat input; B is the constant-external-pressure formula, not the general reversible result. D reverses the maximum-work comparison. “Matched pressures” means the quasistatic limit with an infinitesimal driving difference.

A 忽略了维持等温所需的吸热；B 是恒外压公式；D 把最大功关系说反了。“内外压匹配”指具有无穷小驱动力的准静态极限，并非严格无驱动力时仍持续膨胀。

QUESTION 2 / 26

Free expansion into a vacuum

向真空自由膨胀

A sample of gas expands against an external pressure $p_{ex} = 0$ (free expansion). What is the value of the expansion work w done by the system?

气体对抗 $p_{\mathrm{ex}} = 0$ 的外压膨胀，即自由膨胀。系统的膨胀功 w 是多少？

A. $w = -nRT \ln\left(\frac{V_f}{V_i}\right)$

$w = -nRT \ln\left(\dfrac{V_f}{V_i}\right)$

B. $w = +p_{\mathrm{ex}}\Delta V$

$w = +p_{\mathrm{ex}}\Delta V$

C. $w = -p\Delta V$

$w = -p\Delta V$

D. $w = 0$

$w = 0$

Answer and reasoning / 答案与推导

D

Boundary expansion work is determined by the opposing external pressure, not by the gas's internal pressure. In a vacuum the gas has no external load to displace. This makes the expansion-work integral zero even though its volume increases.

边界膨胀功取决于阻碍膨胀的外压，而不是气体内部压强。向真空膨胀时没有外部负载，因此即使体积增加，膨胀功仍为零。

$$w = -\int_{V_i}^{V_f} p_{\mathrm{ex}}\,\mathrm{d}V,$$
$$p_{\mathrm{ex}} = 0 \quad \Longrightarrow \quad \boxed{w = 0}.$$

Alternatives and conditions / 选项辨析与适用条件

D is the intended answer. However, B also evaluates to zero at the specified $p_{\mathrm{ex}} = 0$, despite having the wrong sign as a general work formula. Thus the options are not numerically unique. Free expansion alone does not imply $q = 0$; an insulated setup adds that condition.

原题预期选 D，但在已给定的 $p_{\mathrm{ex}} = 0$ 下，B 的数值也等于零，只是其一般公式的符号错误。因此本题选项并非严格数值唯一。自由膨胀本身也不等于绝热；只有再加隔热条件才能断言 $q = 0$。

QUESTION 3 / 26

Equivalent pressure and volume formulas

压强式与体积式完全等价

An ideal gas undergoes an isothermal reversible expansion from volume V_1 to volume V_2. Which formula represents the work w performed on the system?

理想气体从 V_1 可逆等温膨胀到 V_2。哪一个公式表示外界对系统所做的功 w？

A. $w = -nRT \ln\left(\frac{P_1}{P_2}\right)$

$$w = -nRT \ln\left(\frac{P_1}{P_2}\right)$$

B. $w = -nRT \ln\left(\frac{V_2}{V_1}\right)$

$$w = -nRT \ln\left(\frac{V_2}{V_1}\right)$$

C. $w = +p_{\text{ex}}(V_2 - V_1)$

$$w = +p_{\text{ex}}(V_2 - V_1)$$

D. $w = -nRT\left(\frac{V_2 - V_1}{V_1}\right)$

$$w = -nRT\left(\frac{V_2 - V_1}{V_1}\right)$$

Answer and reasoning / 答案与推导

B; A also valid

The quiz marks B, but A is exactly equivalent, including its sign. For fixed n, T, Boyle's law gives $P_1V_1 = P_2V_2$. Substitution transforms one logarithm into the other without any approximation.

原题标记 B，但 A 与 B 完全等价，连正负号也一致。固定物质的量和温度时，玻意耳定律给出 $P_1V_1 = P_2V_2$，直接代入即可证明，不涉及近似。

$$\begin{aligned} \frac{V_2}{V_1} &= \frac{P_1}{P_2}, \\ w &= -nRT \ln\left(\frac{V_2}{V_1}\right) \\ &= -nRT \ln\left(\frac{P_1}{P_2}\right) = nRT \ln\left(\frac{P_2}{P_1}\right). \end{aligned}$$

Alternatives and conditions / 选项辨析与适用条件

For expansion $V_2 > V_1$, so all three expressions give $w < 0$. The feedback calling A correct only "in absolute value" is wrong. C has the wrong expansion sign; D is only a small-volume-change approximation when $\ln(1 + x) \simeq x$.

膨胀时 $V_2 > V_1$，上述三式都给出 $w < 0$。原反馈称 A "仅绝对值正确"是错误的。C 的膨胀功符号错误；D 只在体积变化很小时，可作为 $\ln(1 + x) \simeq x$ 的近似。

QUESTION 4 / 26

The chemistry sign convention

化学热力学的能量符号

According to the acquisitive convention used in thermodynamics, what do positive values for heat q and work w signify?

按热力学采用的“系统获得能量为正”的约定，正的热量 q 和功 w 表示什么？

A. Energy is transferred into the system as heat or work, increasing the internal energy of the system.

能量以热或功的形式传入系统，使系统内能增加。

B. Work is done by the system on the surroundings while heat is absorbed by the surroundings.

系统对环境做功，同时环境吸收热量。

C. The internal energy of the system strictly decreases.

系统的内能必然减少。

D. Energy leaves the system to do work on the surroundings or releases heat.

能量离开系统，用于对环境做功或向环境放热。

Answer and reasoning / 答案与推导

A

The acquisitive convention assigns positive signs to energy received by the system. Heat absorbed is positive; work performed on the system is positive. For a closed system with negligible changes in bulk kinetic and potential energies, the first law reads:

“获得为正”的约定从系统角度记账：系统吸热为正，外界对系统做功为正。对封闭系统，忽略整体动能和势能的变化时，热力学第一定律为：

$$\begin{gathered}\Delta U = q + w,\\ q > 0,\ w > 0 \quad\Longrightarrow\quad \Delta U > 0,\\ \delta w_{pV} = -p_{\mathrm{ex}}\,\mathrm{d}V.\end{gathered}$$

Alternatives and conditions / 选项辨析与适用条件

Expansion has $\mathrm{d}V > 0$ and hence $\delta w < 0$; compression has $\delta w > 0$. If one transfer is positive but the other negative, the sign of ΔU depends on their sum. B and D describe energy leaving, while C contradicts two positive inputs.

膨胀时 $\mathrm{d}V > 0$，所以功为负；压缩时功为正。若热和功一正一负，内能是增是减要看总和。B、D 描述能量离开系统；C 则与热和功都为正相矛盾。

QUESTION 5 / 26

State functions and path functions

状态函数与路径函数

Why are internal energy (U) and enthalpy (H) classified as state functions, whereas heat (q) and work (w) are path functions?

为什么内能 U 和焓 H 属于状态函数，而热量 q 与功 w 属于路径函数？

A. Internal energy depends on the history of preparation, while work done is independent of the apparatus used.

内能依赖系统的制备历史，而功与所用装置无关。

B. Values of U and H depend solely on the current state of the system, whereas q and w depend on the specific path taken between states.

U 和 H 只取决于系统当前状态，q 和 w 则取决于两状态间的具体路径。

C. Heat and work are conserved properties of bulk matter, whereas internal energy changes during cyclic processes.

热和功是物质的守恒属性，而内能会在循环后改变。

D. Enthalpy and internal energy can only be defined for open systems, whereas heat and work apply only to closed systems.

焓和内能只能用于开放系统，而热和功只能用于封闭系统。

Answer and reasoning / 答案与推导

B

A complete equilibrium state determines U and H. Their changes depend only on the endpoints. Heat and work instead describe transfers during a chosen process; two paths can divide the same energy change differently between q and w.

完整的平衡状态决定 U 和 H，其变化量只由始末态决定。热和功描述某一过程中的能量传递，同一个内能变化可以通过不同路径，分配为不同的热量和功。

$$\Delta U = U_f - U_i = q_1 + w_1 = q_2 + w_2,$$

$$\oint \mathrm{d}U = 0, \qquad \oint \mathrm{d}H = 0,$$

$$\oint \delta q = -\oint \delta w \quad \text{in a cycle.}$$

Alternatives and conditions / 选项辨析与适用条件

The cycle integrals of heat and work need not vanish individually. A reversibly expanding ideal gas and an insulated freely expanding ideal gas can connect the same isothermal endpoints with different q, w, while both have $\Delta U = 0$. Heat is not a substance stored inside the system.

循环中热和功的积分不必各自为零。理想气体的可逆等温膨胀与绝热自由膨胀，可以连接相同的等温始末态；两者热和功不同，但都满足 $\Delta U = 0$。热不是储存在系统内的一种物质。

QUESTION 6 / 26

Three translational quadratic terms

三个平动二次项

According to the equipartition theorem, what is the translational contribution to the molar internal energy (U_m) of a gas at temperature T?

按能量均分定理，温度为 T 的气体，其摩尔内能 U_{m} 中的平动贡献是多少？

A. $\frac{1}{2}RT$

$\frac{1}{2}RT$

B. $\frac{3}{2}RT$

$\frac{3}{2}RT$

C. $\frac{5}{2}RT$

$\frac{5}{2}RT$

D. $3RT$

$3RT$

Answer and reasoning / 答案与推导

B

Classical equipartition gives $k_{\mathrm{B}}T/2$ per independent quadratic energy term per particle. Translational kinetic energy has three such terms, one for each Cartesian momentum component. Multiplying by Avogadro's constant converts the result to a molar quantity.

经典能量均分定理指出，每个独立能量二次项对每个粒子的平均贡献为 $k_{\mathrm{B}}T/2$。平动动能有三个二次项，分别对应三个笛卡尔动量分量。乘以阿伏伽德罗常数即可得到摩尔量。

$$\epsilon_{\mathrm{trans}} = \frac{p_x^2 + p_y^2 + p_z^2}{2m},$$
$$\langle \epsilon_{\mathrm{trans}} \rangle = \frac{3}{2} k_{\mathrm{B}} T,$$
$$U_{\mathrm{m},trans} = \frac{3}{2} N_{\mathrm{A}} k_{\mathrm{B}} T = \boxed{\frac{3}{2} RT}.$$

Alternatives and conditions / 选项辨析与适用条件

A counts only one spatial direction. C includes two classical rotational terms of a linear molecule; D includes three of a nonlinear molecule. These additional terms are not part of translation. The result requires the classical, nondegenerate regime.

A 只计算了一个空间方向；C 还包括线性分子的两个经典转动项；D 还包括非线性分子的三个转动项。它们不属于平动贡献。本结论要求气体处于经典、非量子简并的适用范围。

QUESTION 7 / 26

When quantum level spacings matter

量子能级间隔何时重要

Under what condition is the equipartition theorem generally unreliable for estimating energy contributions?

在哪一种条件下，能量均分定理通常不能可靠估计能量贡献？

A. Translational motion of gas particles in standard-sized laboratory containers.

气体粒子在通常实验室容器中的平动。

B. Vibrational modes of motion where thermal energy kT is not much greater than the vibrational energy level spacing $\Delta\epsilon$.

热能 kT 并不远大于振动能级间隔 $\Delta\epsilon$ 的振动模式。

C. Rotational motion of heavy linear molecules at room temperature.

重的线性分子在室温下的转动。

D. Monatomic gases undergoing non-relativistic translational motion.

单原子气体的非相对论平动。

Answer and reasoning / 答案与推导

B

Equipartition is a classical approximation. If the thermal energy is not large compared with a mode's quantum spacing, excited states are sparsely populated and the classical contribution is suppressed. Vibrations often have much larger spacings than rotations.

能量均分定理是经典近似。若热能不远大于某运动模式的量子能级间隔，激发态的布居便很少，实际能量与热容贡献低于经典预测。振动能级间隔通常比转动间隔大得多。

$$\Delta\epsilon = \hbar\omega, \qquad x = \frac{\hbar\omega}{k_{\mathrm{B}}T},$$
$$\langle\epsilon_{\mathrm{vib}}\rangle = \frac{\hbar\omega}{2} + \frac{\hbar\omega}{e^{x}-1},$$
$$x \ll 1:\ \text{classical thermal limit}, \qquad x \gg 1:\ \text{excited states suppressed}.$$

Alternatives and conditions / 选项辨析与适用条件

The zero-point term remains even when thermal excitation is frozen out, but it does not contribute to heat capacity if ω is fixed. D is not universally classical: non-relativistic motion alone does not exclude quantum degeneracy at very low temperature or high density.

即使热激发被冻结，零点能仍然存在；若 ω 不变，零点能不贡献热容。此外，D 也不是无条件经典：非相对论并不排除极低温或高密度下的量子简并。

QUESTION 8 / 26

Defining enthalpy

焓的定义及等价恒等式

How is the enthalpy H of a thermodynamic system formally defined in terms of state variables?

热力学系统的焓 H，用状态变量如何正式定义？

A. $H = U + pV$

$H = U + pV$

B. $H = q_V + pV$

$H = q_V + pV$

C. $H = G + TS$

$H = G + TS$

D. $H = U - pV$

$H = U - pV$

Answer and reasoning / 答案与推导

A; C is an identity

The formal definition is $H = U + pV$, so A is the intended choice. The pV term makes enthalpy especially convenient for constant-pressure processes. With pressure-volume work only, constant pressure implies $\Delta H = q_p$.

正式定义为 $H = U + pV$，因此原题预期选 A。加入 pV 项后，焓特别适合处理恒压过程。若只有体积功，恒压时便有 $\Delta H = q_p$。

$$\begin{aligned} H &\equiv U + pV, \\ G \equiv H - TS \quad &\Longrightarrow \quad H = G + TS, \\ \Delta H = \Delta U + p\Delta V &= q_p \quad (p \text{ constant}). \end{aligned}$$

Alternatives and conditions / 选项辨析与适用条件

C is also a true identity once Gibbs energy is defined, although it is not the usual primary definition requested. B confuses the state variable U with a particular path's heat q_V; D has the wrong sign. Thus "which equation is true?" would have more than one correct answer.

在定义了吉布斯能后，C 也是严格正确的恒等式，只是不是本题所问的通常初始定义。B 把状态量 U 与某条路径的热量 q_V 混淆；D 的符号错误。因此若题目改问"哪个等式正确"，便不止一个答案。

QUESTION 9 / 26

Defining constant-volume heat capacity

定容热容的定义

What is the mathematical definition of the constant-volume heat capacity, C_V?

定容热容 C_V 的数学定义是什么？

A. $C_V = \left(\frac{\partial H}{\partial T}\right)_p$

$$C_V = \left(\frac{\partial H}{\partial T}\right)_p$$

B. $C_V = \left(\frac{\partial U}{\partial p}\right)_T$

$$C_V = \left(\frac{\partial U}{\partial p}\right)_T$$

C. $C_V = \left(\frac{\partial U}{\partial T}\right)_V$

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V$$

D. $C_V = \frac{q_p}{\Delta T}$

$$C_V = \frac{q_p}{\Delta T}$$

Answer and reasoning / 答案与推导

C

At fixed volume and composition, C_V is the slope of internal energy with respect to temperature. It may depend on temperature. The subscript on the derivative states what is held fixed, and is part of the definition.

固定体积和组成时，C_V 是内能对温度的变化率，它可以随温度改变。偏导数下标指出保持不变的条件，是定义不可省略的一部分。

$$C_V \equiv \left(\frac{\partial U}{\partial T}\right)_{V,n},$$

$$\Delta U = \int_{T_i}^{T_f} C_V(T)\,\mathrm{d}T,$$

$$C_p \equiv \left(\frac{\partial H}{\partial T}\right)_{p,n}.$$

Alternatives and conditions / 选项辨析与适用条件

A defines C_p instead. B differentiates with respect to pressure, giving the wrong physical quantity and units. D is at best a constant-pressure average heat capacity under additional assumptions, not the definition of C_V. Total heat capacity has units $\mathrm{J\,K^{-1}}$; molar heat capacity has units $\mathrm{J\,mol^{-1}\,K^{-1}}$.

A 定义的是 C_p；B 对压强求导，物理意义和量纲都不对；D 在附加条件下至多是恒压平均热容，不是 C_V 的定义。总热容的单位是 $\mathrm{J\,K^{-1}}$，摩尔热容的单位是 $\mathrm{J\,mol^{-1}\,K^{-1}}$。

QUESTION 10 / 26

Temperature and the Boltzmann population ratio

温度与玻尔兹曼布居比

According to the Boltzmann distribution expression $N_i \propto e^{-\epsilon_i/kT}$, how does increasing the absolute temperature T affect the population of high-energy states?

根据玻尔兹曼分布 $N_i \propto e^{-\epsilon_i/kT}$，提高绝对温度 T 会怎样影响高能态的布居？

A. All molecules fall into the lowest available energy ground state.

所有分子都落入最低能量的基态。

B. Higher energy states become progressively more populated at the expense of lower energy states.

高能态的相对布居逐渐增大，低能态的相对优势减弱。

C. The relative populations of all available energy states become completely independent of temperature.

所有能态的相对布居都完全不再依赖温度。

D. Higher energy states become rapidly depopulated as molecules lose kinetic energy.

分子失去动能，高能态的布居迅速减少。

Answer and reasoning / 答案与推导

B

Compare a higher level j with a lower level i. For fixed energies and degeneracies, their population ratio increases when temperature rises because the energetic penalty becomes smaller relative to $k_{\mathrm{B}}T$.

比较较高能级 j 与较低能级 i。若能级和简并度固定，升温会使高能态相对低能态的布居比增加，因为能量差相对于 $k_{\mathrm{B}}T$ 的代价变小。

$$\frac{N_j}{N_i} = \frac{g_j}{g_i}\exp\left[-\frac{\epsilon_j-\epsilon_i}{k_{\mathrm{B}}T}\right],$$
$$\frac{\mathrm{d}}{\mathrm{d}T}\ln\left(\frac{N_j}{N_i}\right) = \frac{\epsilon_j-\epsilon_i}{k_{\mathrm{B}}T^2} > 0.$$

Alternatives and conditions / 选项辨析与适用条件

Here g_i counts a level's degeneracy; for individual microstates it is one. The rigorous claim is about the high-to-low population ratio. In a many-level spectrum, the absolute population of every excited level need not increase monotonically at all temperatures. A and D describe the opposite tendency; C ignores the exponential temperature dependence.

其中 g_i 表示能级简并度，单个微观态则为一。严格结论是“高能态与低能态的布居比上升”。多能级体系中，并非每个激发能级的绝对布居都在所有温度下单调增加。A、D 把趋势说反；C 忽略了指数中的温度依赖。

QUESTION 11 / 26

Root-mean-square molecular speed

分子的均方根速率

What is the root-mean-square speed (v_{rms}) of molecules of molar mass M in a gas at thermodynamic temperature T according to kinetic molecular theory?

按气体动理论，摩尔质量为 M 的分子在温度 T 下的均方根速率 v_{rms} 是多少？

A. $v_{\mathrm{rms}} = \left(\frac{2RT}{M}\right)^{1/2}$

$$v_{\mathrm{rms}} = \left(\frac{2RT}{M}\right)^{1/2}$$

B. $v_{\mathrm{rms}} = \frac{3kT}{m}$

$$v_{\mathrm{rms}} = \frac{3kT}{m}$$

C. $v_{\mathrm{rms}} = \left(\frac{3RT}{M}\right)^{1/2}$

$$v_{\mathrm{rms}} = \left(\frac{3RT}{M}\right)^{1/2}$$

D. $v_{\mathrm{rms}} = \left(\frac{8RT}{\pi M}\right)^{1/2}$

$$v_{\mathrm{rms}} = \left(\frac{8RT}{\pi M}\right)^{1/2}$$

Answer and reasoning / 答案与推导

C

The mean translational kinetic energy is $3k_{\mathrm{B}}T/2$ per molecule. Root-mean-square speed is defined by the square root of the mean squared speed, not by the ordinary mean speed.

每个分子的平均平动动能为 $3k_{\mathrm{B}}T/2$。均方根速率是“速率平方的平均值再开方”，不是普通平均速率。

$$\frac{1}{2}m\langle v^2\rangle = \frac{3}{2}k_{\mathrm{B}}T,$$

$$v_{\mathrm{rms}} = \sqrt{\langle v^2\rangle} = \boxed{\sqrt{\frac{3RT}{M}}},$$

$$v_{\mathrm{mp}} = \sqrt{\frac{2RT}{M}}, \qquad \langle v\rangle = \sqrt{\frac{8RT}{\pi M}}.$$

Alternatives and conditions / 选项辨析与适用条件

A is the most probable speed; D is the mean speed. B omits the square root and has units of speed squared. In SI calculations use M in $\mathrm{kg\,mol^{-1}}$, not $\mathrm{g\,mol^{-1}}$. For nitrogen at $300\,\mathrm{K}$, $M \simeq 0.0280\,\mathrm{kg\,mol^{-1}}$ gives $v_{\mathrm{rms}} \simeq 517\,\mathrm{m\,s^{-1}}$.

A 是最概然速率，D 是平均速率；B 漏掉平方根，量纲为速率的平方。采用 SI 单位时，摩尔质量必须用 $\mathrm{kg\,mol^{-1}}$。例如 $300\,\mathrm{K}$ 的氮气取 $M \simeq 0.0280\,\mathrm{kg\,mol^{-1}}$，得到约 $517\,\mathrm{m\,s^{-1}}$。

QUESTION 12 / 26

Assumptions of ideal-gas kinetic theory

理想气体动理论的假设

Which of the following is NOT one of the primary foundational assumptions of the kinetic model of ideal gases?

下列哪一项不是理想气体动理论的基本假设？

A. The molecules are in ceaseless random motion obeying classical mechanics.

分子按经典力学规律进行永不停息的随机运动。

B. Collisions between molecules and with container walls are perfectly elastic.

分子之间以及分子与器壁之间的碰撞完全弹性。

C. The gas consists of point-like molecules whose size is negligible relative to the distance between collisions.

分子可视为点粒子，其尺寸相对于碰撞之间的运动距离可忽略。

D. Molecules interact through strong, attractive long-range potential energy fields between collisions.

在碰撞之间，分子通过很强的长程吸引势相互作用。

Answer and reasoning / 答案与推导

D

The classical kinetic model assumes random motion, negligible molecular size, elastic collisions, and no appreciable intermolecular potential energy between collisions. Strong long-range attraction would violate that idealization and alter the pressure-volume relation.

经典气体动理论假设分子随机运动、分子尺寸可忽略、碰撞完全弹性，并忽略碰撞之间的分子间势能。很强的长程吸引作用会破坏这些假设，并改变压强与体积的关系。

$$\begin{aligned} pV &= \frac{1}{3}Nm\langle v^2\rangle, \\ \frac{1}{2}m\langle v^2\rangle &= \frac{3}{2}k_{\mathrm{B}}T, \\ \therefore \quad pV &= Nk_{\mathrm{B}}T = nRT. \end{aligned}$$

Alternatives and conditions / 选项辨析与适用条件

A, B, and C belong to the standard model, so the word NOT makes D the answer. “Elastic” means total kinetic energy is conserved in a collision; it does not mean that each molecule preserves its own speed. The point-particle collision model is an ideal limiting description.

A、B、C 都属于标准假设；题干问“不属于”，因此选 D。“弹性”表示一次碰撞的总动能守恒，并不表示每个分子的速率都不改变。点粒子与碰撞的描述应理解为理想极限模型。

QUESTION 13 / 26

Partial pressure as a composition-weighted pressure

分压的定义与理想气体解释

How is the partial pressure p_J of an individual component J in a gaseous mixture formally defined for any gas (real or ideal)?

对于任意气体混合物，无论真实或理想，组分 J 的分压 p_J 如何正式定义？

A. $p_J = x_J p$, where x_J is the mole fraction of component J and p is total pressure.

$p_J = x_J p$，其中 x_J 是组分 J 的摩尔分数，p 是总压。

B. $p_J = \frac{m_J}{M_J} p$

$$p_J = \frac{m_J}{M_J} p$$

C. $p_J = p - x_J$

$$p_J = p - x_J$$

D. $p_J = \frac{n_J RT}{V_J}$, where V_J is the partial volume.

$p_J = \dfrac{n_J RT}{V_J}$，其中 V_J 是分体积。

Answer and reasoning / 答案与推导

A

In the convention used here, partial pressure is defined as mole fraction times total pressure, including for a real gas mixture. This is distinct from asserting that each real component obeys an independent ideal-gas equation.

按本题采用的约定，分压定义为摩尔分数与总压的乘积，对真实气体混合物也可这样定义。但这不等于宣称每个真实气体组分都独立服从理想气体状态方程。

$$x_J = \frac{n_J}{\sum_K n_K}, \qquad p_J \equiv x_J p,$$

$$\sum_J p_J = p \sum_J x_J = p,$$

$$p_J = \frac{n_J RT}{V} \quad \text{for an ideal-gas mixture.}$$

Alternatives and conditions / 选项辨析与适用条件

The ideal component occupies the full mixture volume V, not a separate V_J as in D. B uses amount of substance rather than a dimensionless fraction; C subtracts unlike units. Real-mixture chemical potentials require fugacity corrections rather than automatically replacing fugacity by $x_J p$.

理想气体组分使用的是混合物的总体积 V，不是 D 中另设的 V_J。B 用了物质的量而非无量纲分数；C 把不同量纲的量相减。真实混合物的化学势通常需要逸度修正，不能自动把逸度等同于 $x_J p$。

QUESTION 14 / 26

SATP and the ideal molar volume

SATP 与理想摩尔体积

What are the standard ambient temperature and pressure (SATP) conditions commonly used to specify gas properties?

通常用于表述气体性质的标准环境温度和压力（SATP）是什么？

A. 298.15 K and 1 bar (10^5 Pa)

298.15 K 和 1 bar（10^5 Pa）。

B. 300 K and 1 atm

300 K 和 1 atm。

C. 273.16 K and 1 bar

273.16 K 和 1 bar。

D. 0 °C (273.15 K) and 1 atm (101.325 kPa)

0 °C（273.15 K）和 1 atm（101.325 kPa）。

Answer and reasoning / 答案与推导

A

Standard ambient temperature and pressure in this quiz means 25 °C and 1 bar. It differs from standard conditions for gases at 273.15 K. At the stated SATP, the ideal molar volume follows directly from $V_\mathrm{m} = RT/p$. [R1, R2]

本题的标准环境温度和压力（SATP）取 25 °C 与 1 bar，不同于采用 273.15 K 的气体标准条件。理想摩尔体积直接由 $V_\mathrm{m} = RT/p$ 求出。[R1, R2]

$$\begin{aligned} T &= 298.15\,\mathrm{K}, \qquad p = 10^5\,\mathrm{Pa}, \\ V_\mathrm{m} &= \frac{(8.314462618)(298.15)}{10^5}\,\mathrm{m^3\,mol^{-1}} \\ &\simeq 24.790\,\mathrm{L\,mol^{-1}}. \end{aligned}$$

Alternatives and conditions / 选项辨析与适用条件

One bar is 100 kPa; one atmosphere is 101.325 kPa. These are different units. D uses an older common STP convention, not SATP. The standard-state superscript alone does not fix the temperature to 298.15 K.

一巴为 100 kPa，一标准大气压为 101.325 kPa，两者不同。D 属于过去常见的一种 STP 约定，不是 SATP。标准态上标本身也不表示温度必定为 298.15 K。

QUESTION 15 / 26

The sign of an exothermic reaction enthalpy

放热反应焓的符号

Which thermodynamic property characterizes an exothermic chemical process occurring at constant pressure?

恒压下发生的放热化学过程具有哪一项热力学特征？

A. The reaction enthalpy is positive ($\Delta_r H > 0$), indicating heat absorption from surroundings.

反应焓为正，$\Delta_r H > 0$，表示从环境吸热。

B. No expansion work can be performed during the reaction.

反应过程中不可能发生膨胀功。

C. The reaction enthalpy is negative ($\Delta_r H < 0$), indicating heat release to surroundings.

反应焓为负，$\Delta_r H < 0$，表示向环境放热。

D. The change in internal energy must be zero ($\Delta U = 0$).

内能变化必为零，即 $\Delta U = 0$。

Answer and reasoning / 答案与推导

C

A constant-pressure reaction releasing heat to its surroundings has $q_{p,\mathrm{sys}} < 0$. For a closed system doing only pressure-volume work, this equals the enthalpy change, so the forward exothermic reaction has negative reaction enthalpy.

恒压反应向环境放热时，系统的 $q_p < 0$。若系统封闭且只做体积功，则恒压热等于焓变，所以正向放热反应的反应焓为负。

$$
\begin{aligned}
q_{p,\mathrm{sys}} &= \Delta H_{\mathrm{sys}} < 0, \\
\Delta H &= \Delta U + \Delta(pV), \\
\Delta_r H &= \Delta_r U + \Delta\nu_{\mathrm{g}} RT \quad \text{(ideal gases; condensed } pV \text{ neglected).}
\end{aligned}
$$

Alternatives and conditions / 选项辨析与适用条件

A describes an endothermic reaction. Exothermicity does not prohibit expansion work and does not imply $\Delta U = 0$, so B and D do not follow. If the reaction is reversed, the sign of its reaction enthalpy reverses. Additional non-expansion work requires extra care when identifying heat with ΔH.

A 描述吸热反应。放热并不禁止膨胀功，也不推出 $\Delta U = 0$，因此 B、D 不成立。反应方向反转时，反应焓符号也反转。若存在额外的非体积功，则不能直接把热量等同于焓变。

QUESTION 16 / 26

Kirchhoff temperature correction

基尔霍夫温度修正

How does Kirchhoff's law calculate the standard reaction enthalpy $\Delta_r H^{\ominus}$ at temperature T' using data from temperature T?

基尔霍夫定律如何用温度 T 下的数据计算温度 T' 下的标准反应焓？

A. $\Delta_r H^{\ominus}(T') = \Delta_r U^{\ominus}(T) + nRT'$

$\Delta_r H^{\circ}(T') = \Delta_r U^{\circ}(T) + nRT'$

B. $\Delta_r H^{\ominus}(T') = \Delta_r H^{\ominus}(T) \times \frac{T'}{T}$

$\Delta_r H^{\circ}(T') = \Delta_r H^{\circ}(T)\dfrac{T'}{T}$

C. $\Delta_r H^{\ominus}(T') = \Delta_r H^{\ominus}(T) - \Delta C_V^{\ominus} \times (T' - T)$

$\Delta_r H^{\circ}(T') = \Delta_r H^{\circ}(T) - \Delta C_V^{\circ}(T' - T)$

D. $\Delta_r H^{\ominus}(T') = \Delta_r H^{\ominus}(T) + \Delta C_p^{\ominus} \times (T' - T)$

$\Delta_r H^{\circ}(T') = \Delta_r H^{\circ}(T) + \Delta C_p^{\circ}(T' - T)$

Answer and reasoning / 答案与推导

D, with constant heat-capacity difference

Reaction enthalpy changes with temperature because products and reactants have different heat capacities. Kirchhoff's law differentiates the stoichiometric sum of molar enthalpies. The linear formula in D requires an approximately constant reaction heat-capacity difference over the interval.

产物与反应物热容不同，所以反应焓随温度改变。基尔霍夫定律来自对摩尔焓的化学计量加和求导。D 的线性式要求在所考察温区内，反应热容差近似为常数。

$$\Delta_r C_p^{\circ}(T) = \sum_i \nu_i C_{p,\mathrm{m},i}^{\circ}(T),$$

$$\begin{aligned}\Delta_r H^{\circ}(T') &= \Delta_r H^{\circ}(T) + \int_T^{T'} \Delta_r C_p^{\circ}(\tau)\,\mathrm{d}\tau, \\ &\simeq \Delta_r H^{\circ}(T) + \Delta_r C_p^{\circ}(T' - T).\end{aligned}$$

Alternatives and conditions / 选项辨析与适用条件

Use positive stoichiometric coefficients for products and negative ones for reactants. C uses the wrong heat capacity and sign. A confuses an internal-energy conversion with a temperature correction; B assumes an unjustified proportionality. If a phase transition intervenes, split the heating path and include latent enthalpy.

化学计量数对产物取正、对反应物取负。C 的热容种类和符号都不对；A 混淆了内能换算与温度修正；B 假设了没有依据的比例关系。若跨越相变，应分段积分并加入相变焓。

QUESTION 17 / 26

Adiabatic does not mean isothermal

绝热不等于等温

What occurs when a boundary dividing a system and surroundings is described as adiabatic?

系统与环境之间的边界若为绝热边界，意味着什么？

A. Energy easily passes through as heat due to temperature differences.

由于温差，能量很容易以热的形式穿过边界。

B. Matter passes freely while heat exchange is prohibited.

物质可以自由通过，但禁止热交换。

C. No transfer of energy as heat can take place through the boundary ($q = 0$).

能量不能以热的形式通过边界，即 $q = 0$。

D. The temperature of the system must remain strictly constant throughout any process.

任何过程中系统温度都必须严格不变。

Answer and reasoning / 答案与推导

C

An adiabatic boundary blocks heat transfer. It need not block work, and it does not require constant temperature. In a closed adiabatic system, work can change the internal energy and therefore the temperature.

绝热边界阻止热传递，但不一定阻止功，也不要求温度不变。对封闭绝热系统，功仍可改变内能，进而改变温度。

$$
\begin{aligned}
q = 0, \qquad & \Delta U = w, \\
nC_{V,\mathrm{m}}(T_f - T_i) = w \quad & \text{if } C_{V,\mathrm{m}} \text{ is constant}, \\
TV^{\gamma-1} = \text{constant} \quad & \text{(reversible ideal-gas adiabat)}.
\end{aligned}
$$

Alternatives and conditions / 选项辨析与适用条件

Compression typically raises an insulated gas's temperature; expansion can lower it. A describes a heat-conducting boundary. B adds an unsupported statement about matter transfer. D confuses adiabatic with isothermal. The power-law relation additionally needs constant heat capacities and $\gamma = C_p/C_V$.

隔热气体被压缩时通常升温，膨胀时可以降温。A 描述导热边界；B 对物质能否通过作了题干未给出的附加判断；D 混淆绝热和等温。最后的幂律还要求热容近似恒定，且 $\gamma = C_p/C_V$。

QUESTION 18 / 26

A bomb calorimeter measures constant-volume heat

氧弹量热计测量定容热

Which laboratory apparatus measures ΔU directly for a combustion reaction by ensuring zero expansion work is done?

哪一种实验装置通过使膨胀功为零，直接测量燃烧反应的 ΔU？

A. A capacitance manometer.

电容式压力计。

B. A constant-volume bomb calorimeter.

定容氧弹量热计。

C. An adiabatic flame calorimeter operating at constant pressure.

在恒压下工作的绝热火焰量热计。

D. An open isobaric calorimeter vessel.

开放的恒压量热容器。

Answer and reasoning / 答案与推导

B

A rigid bomb keeps the reacting system at constant volume, so pressure-volume boundary work vanishes. After correcting for ignition and other contributions, the reaction heat at fixed volume gives its internal-energy change.

刚性氧弹使反应系统保持定容，因此边界体积功为零。扣除点火等额外贡献后，定容反应热给出反应的内能变化。

$$\begin{aligned} \Delta V = 0 \quad &\Longrightarrow \quad w_{pV} = 0, \\ q_{V,\mathrm{rxn}} &= \Delta U_{\mathrm{rxn}}, \\ q_{\mathrm{cal}} = C_{\mathrm{cal}}\Delta T, \qquad &\Delta U_{\mathrm{rxn}} \simeq -C_{\mathrm{cal}}\Delta T. \end{aligned}$$

Alternatives and conditions / 选项辨析与适用条件

The calorimeter's measured temperature rise must be converted using a calibrated heat capacity; ΔU is inferred, not read from a direct energy dial. A measures pressure. C and D are constant-pressure arrangements, associated with enthalpy under the usual work assumptions. Correct reported energies to the specified initial and final states.

装置实际测量温升，再用标定热容换算，因此“直接测量”是指对应内能而非焓，并不是直接读出能量。A 测压；C、D 为恒压装置，在通常做功条件下对应焓变。最终数值还应修正到指定始末态。

QUESTION 19 / 26

Helmholtz energy and maximum total work

亥姆霍兹能与最大总输出功

At constant temperature, what does the change in Helmholtz energy (ΔA) represent for a system undergoing a state change?

恒温条件下，系统状态改变时的亥姆霍兹能变化 ΔA 对应哪一种功？

A. The maximum non-expansion work extractable at constant pressure.

恒压下可提取的最大非膨胀功。

B. The heat transferred during an irreversible constant-pressure process.

不可逆恒压过程中传递的热量。

C. The maximum total work ($w_{\max}$) extractable from the system.

可从系统提取的最大总功 $w_{\max}$。

D. The energy lost strictly due to friction and irreversible viscous drag.

仅由摩擦和不可逆黏滞阻力损失的能量。

Answer and reasoning / 答案与推导

C, with a sign correction

For a closed system exchanging heat with a reservoir at fixed T, combine the first law with $\Delta S \geq q/T$. This bounds total work, including expansion and non-expansion contributions. Equality is attained in the reversible limit.

封闭系统与温度固定的热库换热时，将第一定律与 $\Delta S \geq q/T$ 联立，可得到总功的界限。总功包括体积功和非体积功，可逆极限时取等号。

$$
\begin{aligned}
q &\leq T\Delta S, \\
w_{\text{on}} = \Delta U - q &\geq \Delta U - T\Delta S = \Delta A, \\
W_{\text{out}} = -w_{\text{on}} &\leq -\Delta A, \\
\boxed{W_{\text{out,max}} = -\Delta A}, &\qquad w_{\text{on,rev}} = \Delta A.
\end{aligned}
$$

Alternatives and conditions / 选项辨析与适用条件

C identifies the correct kind of work, but "extractable work" as a positive output is $-\Delta A$, not ΔA. A describes the Gibbs-energy result. B is a heat quantity; D describes dissipation rather than the available reversible work.

C 指出了正确的功的种类，但若"可提取功"按正的输出量记，应为 $-\Delta A$，不是 ΔA。A 对应吉布斯能；B 是热量；D 指耗散，而不是可逆过程可获得的功。

QUESTION 20 / 26

Gibbs energy and maximum non-expansion work

吉布斯能与最大非体积功

How does the change in Gibbs energy (ΔG) relate to work for a process occurring at constant temperature and constant pressure?

恒温恒压过程中，吉布斯能变化 ΔG 与功有何关系？

A. ΔG equals the total expansion work done against the atmosphere.

ΔG 等于对抗大气所做的全部膨胀功。

B. ΔG represents total heat absorbed by the system during free expansion.

ΔG 表示自由膨胀过程中系统吸收的总热量。

C. ΔG is strictly equal to zero for all spontaneous chemical processes.

所有自发化学过程的 ΔG 都严格为零。

D. ΔG equals the maximum non-expansion (additional) work extractable from the system.

ΔG 等于系统可输出的最大非膨胀功（附加功）。

Answer and reasoning / 答案与推导

D, with a sign correction

At constant T, p, subtract the unavoidable pressure-volume work from the total work balance. The Gibbs-energy decrease then bounds the useful additional work, such as reversible electrical work from a cell.

恒温恒压时，从总功中扣除压力—体积功，吉布斯能的减少量便限制可获得的附加功，例如电池的可逆电功。

$$
\begin{aligned}
w_{\mathrm{on}} &= -p\Delta V + w_{\mathrm{nonPV,on}},\\
\Delta G &= \Delta U + p\Delta V - T\Delta S,\\
w_{\mathrm{nonPV,on}} &\geq \Delta G,\\
\boxed{W_{\mathrm{nonPV,out,max}} = -\Delta G}&.
\end{aligned}
$$

Alternatives and conditions / 选项辨析与适用条件

D is the intended category, but positive extracted work equals the decrease $-\Delta G$. If signed work on the system is used, its reversible value is ΔG. A confuses non-expansion and expansion work. B has no general basis. C confuses spontaneous change, $\Delta G < 0$ under the usual constraints, with equilibrium.

D 的功的类别正确，但按正值记录的输出功等于 $-\Delta G$。若采用外界对系统做功的带符号量，可逆值才等于 ΔG。A 混淆了非体积功与体积功；B 没有一般依据；C 则把通常约束下 $\Delta G < 0$ 的自发变化与平衡混淆。

QUESTION 21 / 26

Perfect gas versus ideal mixing

理想气体与理想混合

What key structural distinction exists between a 'perfect gas' and an 'ideal mixture' according to physical chemistry conventions?

按物理化学中的常见约定，“理想气体”和“理想混合物”的关键结构差异是什么？

A. A perfect gas obeys classical mechanics, whereas ideal mixture components obey quantum mechanics.

理想气体服从经典力学，而理想混合物的组分服从量子力学。

B. Ideal mixtures obey Boyle's law exactly at all pressures, while perfect gases obey it only as $p \to 0$.

理想混合物在任何压强都严格服从玻意耳定律，而理想气体只在零压极限服从。

C. Intermolecular interactions are completely zero in a perfect gas, whereas in an ideal mixture interactions are identical but non-zero.

理想气体中的分子间相互作用为零，而理想混合物中的相互作用相同但不为零。

D. A perfect gas consists of charged ions, while ideal mixtures contain only neutral atoms.

理想气体由带电离子组成，而理想混合物只含中性原子。

Answer and reasoning / 答案与推导

C, as the intended simplified model

A perfect gas neglects intermolecular forces and molecular excluded volume. An ideal solution or mixture need not lack interactions: mixing unlike molecules can be energetically equivalent to mixing like molecules. Ideal mixing describes a thermodynamic mixing behavior, not a universal requirement for nonzero forces.

理想气体忽略分子间作用和分子排除体积。理想溶液或混合物则可以存在相互作用；不同种分子混合时的作用，在能量上可与同种分子作用等效。“理想混合”描述热力学混合行为，并不普遍要求作用力非零。

$$\Delta_{\mathrm{mix}} H = 0, \qquad \Delta_{\mathrm{mix}} V = 0,$$
$$\Delta_{\mathrm{mix}} S = -nR \sum_i x_i \ln x_i,$$
$$\mu_i = \mu_i^*(T,p) + RT \ln x_i \quad \text{(ideal solution)}.$$

Alternatives and conditions / 选项辨析与适用条件

C is the closest intended option, but its phrase “non-zero” is not a necessary condition. A mixture of perfect gases is itself an ideal mixture with negligible interactions. The “equal interactions” picture is a useful sufficient molecular model, not the most general definition.

C 最接近出题意图，但“非零”不是必要条件。多种理想气体的混合物本身也是相互作用可忽略的理想混合物。“各类相互作用等同”是有用的充分分子模型，而不是最一般的定义。

QUESTION 22 / 26

Ideal-gas internal energy is independent of volume

理想气体内能与体积无关

How does the internal energy U of a sample of a perfect gas depend on the volume V it occupies at constant temperature?

恒温下，一定量理想气体的内能 U 如何依赖其体积 V？

A. Internal energy decreases rapidly as volume increases because collision frequency drops.

体积增大使碰撞频率降低，因此内能迅速减少。

B. Internal energy is completely independent of volume because there are no intermolecular forces between particles.

由于不存在分子间作用力，内能完全不依赖体积。

C. Internal energy increases linearly as volume expands due to increased translational space.

由于可用平动空间增加，内能随体积线性增加。

D. Internal energy is inversely proportional to V^2 according to kinetic theory.

按动理论，内能与 V^2 成反比。

Answer and reasoning / 答案与推导

B

For a fixed amount and composition of perfect gas, intermolecular potential energy is absent. At a given temperature, increasing the volume changes the number of translational states and the entropy, but not the classical mean translational kinetic energy.

对物质的量和组成固定的理想气体，不存在分子间势能。固定温度时，增大体积会改变平动状态数与熵，却不会改变经典平均平动动能。

$$\left(\frac{\partial U}{\partial V}\right)_T = T\left(\frac{\partial p}{\partial T}\right)_V - p,$$
$$p = \frac{nRT}{V} \quad \Longrightarrow \quad \left(\frac{\partial U}{\partial V}\right)_T = 0,$$
$$U = U(T) \quad \text{at fixed amount and composition.}$$

Alternatives and conditions / 选项辨析与适用条件

A changes collision frequency without demonstrating an energy change; C confuses state count with mean energy. D misuses confinement-level scaling as a bulk classical-gas energy law. Monatomic $U = 3nRT/2$ is a special case; molecular ideal gases can also have rotational and vibrational contributions depending on T.

A 只说明碰撞频率变化，不能推出内能变化；C 混淆状态数与平均能量；D 把受限粒子的能级标度误当作宏观经典气体定律。单原子气体的 $U = 3nRT/2$ 是特例，分子气体还可有随温度改变的转动和振动贡献。

QUESTION 23 / 26

Converting Celsius to kelvin

摄氏度与开尔文换算

Which relation converts a Celsius temperature θ (°C) to its equivalent thermodynamic temperature T (K)?

摄氏温度 θ 与热力学温度 T 之间的换算式是哪一个？

A. $T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$

$T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.15$

B. $T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.16$

$T/\mathrm{K} = \theta/^\circ\mathrm{C} + 273.16$

C. $T/\mathrm{K} = \frac{5}{9}(\theta/^\circ\mathrm{C} - 32)$

$T/\mathrm{K} = \dfrac{5}{9}(\theta/^\circ\mathrm{C} - 32)$

D. $T/\mathrm{K} = \theta/^\circ\mathrm{C} - 273.15$

$T/\mathrm{K} = \theta/^\circ\mathrm{C} - 273.15$

Answer and reasoning / 答案与推导

A

The Celsius and kelvin scales have equal-sized temperature intervals but different zero points. A Celsius temperature of zero corresponds exactly to 273.15 K. The equation uses numerical values divided by their units.

摄氏温标与开尔文温标的温度间隔大小相同，但零点不同。零摄氏度恰好对应 273.15 K。公式中“温度除以单位”表示该单位下的数值。

$$\boxed{\frac{T}{\mathrm{K}} = \frac{\theta}{^\circ\mathrm{C}} + 273.15},$$

$$25\,^\circ\mathrm{C} \longleftrightarrow 298.15\,\mathrm{K},$$

$$\Delta T = 10\,\mathrm{K} \longleftrightarrow \Delta\theta = 10\,^\circ\mathrm{C}.$$

Alternatives and conditions / 选项辨析与适用条件

B confuses the offset with the familiar water triple-point temperature. C is the Fahrenheit-to-Celsius conversion applied to the wrong variable; D uses the wrong sign. A temperature difference needs no 273.15 offset, whereas $pV = nRT$ and Boltzmann factors require absolute temperature.

B 把换算偏移量与熟悉的水三相点温度混淆；C 是错误套用华氏转摄氏关系；D 的符号相反。温差换算不加 273.15，但理想气体方程和玻尔兹曼因子必须使用绝对温度。

QUESTION 24 / 26

Work and heat are modes of energy transfer

功和热是能量传递方式

From a physical, molecular standpoint, how is work fundamentally distinguished from heat during energy transfer into surroundings?

从分子层面的物理图像看，向环境传递能量时，功与热的根本区别是什么？

A. Work is energy transfer that utilizes uniform motion of atoms in surroundings, whereas heat utilizes random thermal motion.

功对应环境中原子的有序一致运动，而热对应随机热运动。

B. Heat involves net directional displacement of masses, whereas work involves chaotic motion.

热涉及物体的净定向位移，而功涉及杂乱运动。

C. Heat requires macroscopic mechanical pistons, whereas work occurs solely via electromagnetic radiation.

传热必须使用宏观活塞，而做功只能通过电磁辐射实现。

D. Work can only transfer potential energy, whereas heat transfers exclusively rotational energy.

功只能传递势能，而热只能传递转动能。

Answer and reasoning / 答案与推导

A

The quiz uses the molecular picture of ordered motion for work and random thermal motion for heat. More generally, heat is energy transfer caused by a temperature difference; work is transfer through other generalized force-displacement mechanisms.

本题以“有序运动对应功，随机热运动对应热”作为分子图像。更一般地说，热是因温差发生的能量传递；功则通过其他广义力—位移机制传递能量。

$$\begin{aligned} \delta w_{\mathrm{mech,on}} &= F_{\mathrm{ext}}\,\mathrm{d}x, \\ \delta w_{pV,\mathrm{on}} &= -p_{\mathrm{ex}}\,\mathrm{d}V, \\ \Delta U &= q + w. \end{aligned}$$

Alternatives and conditions / 选项辨析与适用条件

A is the best elementary description, not a claim that every kind of work literally translates all atoms uniformly. Electrical and magnetic work also exist. Friction can convert supplied work into random internal motion, so the final motion alone does not determine whether energy entered as heat or work.

A 是最合适的初步描述，并不是说所有形式的功都让全部原子整齐平移，电功、磁功同样存在。摩擦也可把输入的功耗散为随机内运动，因此不能仅凭最终的分子运动来判断能量最初以热还是功进入。

QUESTION 25 / 26

Mechanical balance at a movable wall

可移动隔板的机械平衡

What constitutes a state of mechanical equilibrium between two gases sharing a common movable wall?

两份气体之间由一块可移动壁隔开。达到机械平衡的条件是什么？

A. The mole fractions of both gas samples become equal to unity.

两份气体的摩尔分数都等于一。

B. The pressures exerted on either side of the movable wall are identical ($p_1 = p_2$).

可移动壁两侧压强相同，即 $p_1 = p_2$。

C. The volumes of the two containers are exactly equal regardless of particle count.

无论粒子数多少，两边容器体积都必须相等。

D. The temperatures of the two gases are equal while their pressures differ greatly.

两份气体温度相等，但压强可以相差很大。

Answer and reasoning / 答案与推导

B

For a frictionless movable wall of equal area on each side, with no weight or other external load along its motion, mechanical equilibrium requires the opposing pressure forces to balance. That gives equal pressures, not necessarily equal volumes or temperatures.

若可移动隔板无摩擦、两侧受压面积相同，且运动方向上没有重力或额外负载，机械平衡要求两侧压力平衡，因此压强相等，但体积和温度未必相等。

$$F_{\text{net}} = (p_1 - p_2)A = 0,$$

$$\therefore \quad \boxed{p_1 = p_2}.$$

$$p_1 = p_2,\ T_1 = T_2 \quad \Longrightarrow \quad \frac{V_1}{V_2} = \frac{n_1}{n_2} \quad \text{(ideal gases)}.$$

Alternatives and conditions / 选项辨析与适用条件

The second relation illustrates why equal pressures do not require equal volumes. Thermal equilibrium concerns temperature, and chemical equilibrium concerns chemical potentials. If the wall is weighted, sprung, or subject to friction, include those forces; pressure equality alone is then not the general balance condition.

第二个关系说明等压并不要求等体积。热平衡涉及温度，化学平衡涉及化学势。若隔板有重物、弹簧或摩擦，还必须加入相应力，此时压强相等不再是最一般的平衡条件。

QUESTION 26 / 26

The dilute-gas limit

低密度气体极限

Why does a real gas behave increasingly like a perfect gas as the pressure approaches zero ($p \to 0$)?

当压强趋于零（$p \to 0$）时，真实气体为什么越来越接近理想气体？

A. The temperature automatically increases to infinity as pressure falls.

压强降低时，温度自动升到无穷大。

B. The average distance between molecules becomes so large that intermolecular interactions and molecular volume become negligible.

分子平均间距增大，使分子间相互作用与分子自身占据体积的影响可以忽略。

C. Molecular speeds decrease to zero, stopping all collisions with container walls.

分子速率降为零，不再与器壁碰撞。

D. The molar mass of the gas particles decreases proportionally with pressure.

气体分子的摩尔质量随压强成比例下降。

Answer and reasoning / 答案与推导

B

At fixed nonzero temperature for an ordinary neutral gas, lowering pressure reduces density and increases typical intermolecular separation. Both excluded-volume and interaction corrections become small compared with the ideal-gas contribution.

对普通中性气体，固定非零温度并降低压强，会降低数密度、增加平均分子间距。排除体积与分子间相互作用所造成的修正，相对于理想气体项逐渐变小。

$$Z \equiv \frac{pV_{\mathrm{m}}}{RT},$$

$$Z = 1 + \frac{B(T)}{V_{\mathrm{m}}} + \frac{C(T)}{V_{\mathrm{m}}^2} + \cdots,$$

$$p \to 0 \text{ at fixed } T: \quad V_{\mathrm{m}} \to \infty, \qquad \boxed{Z \to 1}.$$

Alternatives and conditions / 选项辨析与适用条件

The molecular speed remains set mainly by temperature; it does not approach zero simply because pressure falls. Temperature does not automatically diverge, and molar mass does not change. The fixed-temperature dilute limit matters: lowering temperature simultaneously can introduce condensation or quantum effects outside this simple argument.

分子速率主要由温度决定，不会仅因压强下降就趋于零；温度不会自动发散，摩尔质量也不改变。这里强调固定温度的稀薄极限；若同时降温，可能出现凝结或量子效应，超出本题的简单论证。

Formula map / 核心公式汇总

$$
\begin{aligned}
\Delta U &= q + w, \qquad H = U + pV,\\
w_{pV} &= -\int p_{\mathrm{ex}}\,\mathrm{d}V,\\
w_{\mathrm{rev},T} &= -nRT\ln\!\left(\frac{V_f}{V_i}\right),\\
C_V &= \left(\frac{\partial U}{\partial T}\right)_V, \qquad C_p = \left(\frac{\partial H}{\partial T}\right)_p,\\
U_{\mathrm{m},trans} &= \frac{3}{2}RT, \qquad v_{\mathrm{rms}} = \sqrt{\frac{3RT}{M}},\\
p_J &= x_J p, \qquad Z = \frac{pV_{\mathrm{m}}}{RT},\\
\Delta_r H^\circ(T') &= \Delta_r H^\circ(T) + \int_T^{T'} \Delta_r C_p^\circ(\tau)\,\mathrm{d}\tau,\\
W_{\mathrm{out,max}} &= -\Delta A \quad (T \text{ fixed}),\\
W_{\mathrm{nonPV,out,max}} &= -\Delta G \quad (T, p \text{ fixed}).
\end{aligned}
$$

The reversible-work and molecular-speed formulas use the classical ideal-gas model. Maximum-work equalities are reversible limits under the reservoir and boundary-work assumptions explained in Questions 19 and 20.

可逆膨胀功与分子速率公式采用经典理想气体模型；最大功等式是第 19、20 题所述热库与边界做功条件下的可逆极限。

Sources / 来源

Primary question source: the visible Notebook quiz *2 Quiz heat-work multiple-choice*, read on 29 October 2026. The seven source PDFs listed by the notebook were not separately read. Translations, derivations, examples, and editorial corrections were developed for this guide.

题目主来源为当前 Notebook 中的“2 Quiz heat-work multiple-choice”，读取日期为 2026 年 10 月 29 日。未单独读取其列出的七份来源 PDF。翻译、推导、例子和修正为本讲义整理。

https://notebook.google.com/notebook/982933e4-76f6-431d-b28f-bc67d4369556

[**R1**] University of Texas at Austin, Chemistry: “Standard Conditions” (SATP convention).
https://ch301.cm.utexas.edu/gases/ideal-gas-law/standard-conditions.html

[**R2**] IUPAC Gold Book: “Standard conditions for gases” and “Standard atmosphere” (distinguishing 1 bar from 1 atm).
https://goldbook.iupac.org/terms/view/S05910
https://goldbook.iupac.org/terms/view/S05906